FIBRE PRODUCTS OF UNIVERSAL THICKENINGS STAY FLAT OVER THE TARGET WHEN THE BASE THICKENING IS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3280-3293 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3295-3306, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings, formally unramified morphisms of algebraic spaces, and preservation of flatness under base change; the comparison is retained.

Source reference. Original lemma lemma-conormal-functorial-flat: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 557–570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-fibre-product: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3251–3267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-flat: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3306–3306. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} Z & \xrightarrow{h} & X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ W & \xrightarrow{h'} & Y \end{array}$$

be a fibre product diagram of algebraic spaces over S with h' formally unramified and g flat. In this case the corresponding map $Z' \to W'$ of universal first order thickenings is flat, and $f^*\mathcal{C}_{W/Y} \to \mathcal{C}_{Z/X}$ is an isomorphism.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Flatness is preserved under base change, see Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-flat. Hence the first statement follows from the description of W' in Lemma lemma-universal-thickening-fibre-product. It is clear that $X \times_Y W'$ is a first order thickening. It is straightforward to check that it has the universal property because W' has the universal property (by mapping properties of fibre products). See Lemma lemma-conormal-functorial-flat for why this implies that the map of conormal sheaves is an isomorphism. $\square$